

CSCE 632 ACCESSIBLE COMPUTING · MODULE 04 · LECTURE NOTES

Hearing and Speech

Audio alternatives, and not requiring a voice.

MODULE OUTCOMES

- State the requirements for audio content.
- Distinguish captions, subtitles, and transcripts.
- Explain audio description and when it is needed.
- Explain why voice-only input excludes people.
- Test an interface with the sound off.

1 The requirement

NO INFORMATION MAY BE AVAILABLE ONLY AS AUDIO, AND NONE ONLY AS VIDEO

Any information carried by sound needs a text or visual equivalent — speech, alarms, status tones, error beeps, and the entire audio track of any video — which is the half of the requirement everybody knows.

And symmetrically, any information carried only visually within a video needs an audio or text equivalent — which is audio description (§3) and is the half people forget, including in otherwise careful organisations.

So the test is: turn the sound off, then look away — and anything you lose in either direction is a defect — which is about two minutes of work and finds essentially everything in this module.

Plus the situational case here is enormous — most video on the internet is watched without sound, in feeds, in offices, and on public transport — which makes this the clearest curb cut in the course (Module 02 §1) and the easiest requirement to get agreement on.

2 Captions and transcripts

The thing	What it actually is	What it is for
Captions	Synchronised text of all audio content, including speaker identification and significant non-speech sound.	Watching without hearing — the accessibility requirement.

The thing	What it actually is	What it is for
Subtitles	Synchronised translation of the spoken dialogue only.	Understanding a different language. Not an accessibility feature — see the note.
Transcript	The full text of the audio, unsynchronised, as a document.	Reading instead of watching, searching, skimming, and quoting — and frequently what people prefer.
Audio description	Narration of the essential visual content, in the gaps or as an extended track.	Watching without seeing (§3).
Sign language interpretation	A signed track, by a human interpreter.	Users whose first language is a signed one; a AAA-level criterion.

Subtitles are not captions — they omit speaker identification and non-speech audio, both of which are information (who is talking; that an alarm is sounding; that the line went dead) — and a player that labels the two interchangeably is a common and consequential defect, because a user selecting 'subtitles' believing them to be captions gets a silently degraded experience. The subtitles-are-not-captions distinction is the one most often got wrong, including in procurement documents.

AND AUTOMATIC CAPTIONS ARE A DRAFT, NOT A DELIVERABLE

Automatic speech recognition has improved enormously and still fails on exactly the hard parts — proper names, technical terminology, strong or less-represented accents, overlapping speakers, and punctuation.

Which means the errors are concentrated in the content words — precisely the terms carrying the meaning — and a caption track with the technical vocabulary wrong is worse than useless for a lecture, because it is confidently incorrect rather than obviously absent.

And they do not caption non-speech audio at all: no speaker identification, no '[alarm sounds]', no '[laughter]', no '[the line disconnects]' — all of which is information and some of which is the point of the scene.

So: generate automatically and correct manually — which is far cheaper than transcribing from nothing, takes a fraction of the media's duration, and is the honest workflow rather than a compromise (Module 12 §3's cost argument).

3 Audio description

- **Whenever essential information appears visually without being spoken — on-screen text, a diagram, a demonstration, a pointed-at thing, a gesture, or a facial reaction that carries the meaning.**
- **Which is most instructional and demonstration video — 'as you can see here' followed by silence is the characteristic failure, and it is extremely common** in exactly the material students are asked to learn from.
- **And the cheapest fix is to not need it: narrate what you are doing as you do it** — 'I'm opening the settings panel, and the third option is the one we want' — **which produces a video that needs no separate description track and is better anyway**, including for sighted viewers who looked away.
- **Which is the design-for-the-constraint move in its purest form** (Module 01 §2) — **the accessible version is the better version here, not an addition to it**, and that is the strongest form the argument takes.
- **And a text transcript with the visual content described in it is frequently sufficient** for conformance and in practice, **and is much cheaper than a described audio track** — which matters when the alternative is nothing at all.

4 Speech as input

A VOICE-ONLY INTERFACE EXCLUDES ANYBODY WHO DOES NOT SPEAK THE WAY IT EXPECTS

Non-speaking people, people with dysarthria, a stammer, or a tracheostomy, people with atypical or less-represented accents, and people who simply cannot speak aloud right now are all excluded by a voice-only path — which is a wider group than the design usually assumes.

And speech recognition accuracy varies systematically by accent and by speech pattern — which is CSCE 638 Module 12's measured disparity arriving here as an access requirement rather than as a fairness finding, and is the same fact either way.

So voice must be an alternative input and never the only one — a text path, a keypad path, or a human path has to exist — which is a specific and frequently violated requirement for telephone systems, voice assistants, and in-car interfaces.

And the situational case is universal: a quiet office, a meeting, a sleeping child, a crowded train, a sore throat — which makes this another case where the requirement needs no charity to be obvious, only the observation that nobody wants to talk to their phone in a meeting.

AND THE TESTS, WHICH TAKE FIVE MINUTES

Mute everything and use the interface normally — and list every notification, alarm, error, and status tone you missed, which is usually more than you expected.

Play every video with the sound off and read the captions — checking for speaker identification and non-speech audio, not only for word accuracy, since those are what automatic captions omit entirely (§2).

Then play it with the screen off and listen — which is the audio description test and the one nobody runs, and which finds the 'as you can see here' failures immediately.

And find every path that requires speaking, and check that a non-speech alternative exists. **Five minutes, four tests** — and **this module's requirements are the cheapest to meet of any in the course**, which makes their frequent absence harder to explain than most.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
The WAI media accessibility guide (free)	The whole module , free and well organised — with the captions, transcripts, and description requirements separated properly.
Understanding WCAG 1.2.x on time-based media (free)	§§1 to 3's criteria , with the levels and the exceptions explained.
The DCMP Captioning Key (free)	§2 in practice — how to caption well, including placement, timing, and non-speech conventions.
Koenecke et al. — Racial disparities in automated speech recognition (free)	§4's disparity, measured — and the same study CSCE 638 Module 12 uses.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Mute your machine for an hour** and list what you missed.
- 2 State the audio requirement in both directions** and give an example of each failure.
- 3 Compare a subtitle track and a caption track** for the same video.
- 4 Generate automatic captions** for a technical talk and count the content-word errors.
- 5 Correct them**, and record how long it took per minute of media.

- 6 **Find three instructional videos** with 'as you can see here' failures.
- 7 **Record a two-minute demonstration** narrating everything you do.
- 8 **Write a transcript with the visual content described.**
- 9 **Find a voice-only path** in a system you use, and look for the alternative.
- 10 **Run all four tests from the closing callout** on your own project.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the requirement in both directions.
- 2 Which half is forgotten, and what is it called?
- 3 Give the five audio alternatives and what each is for.
- 4 Why are subtitles not captions?
- 5 Why do automatic caption errors concentrate in content words?
- 6 What do automatic captions omit entirely?
- 7 When is audio description needed, and what is the cheapest fix?
- 8 Why is that fix the design-for-the-constraint move?
- 9 Who does a voice-only interface exclude?
- 10 Give the four tests and what each finds.